llmXive Automated Review of SANA-WM: Efficient Minute-Scale World Modeling with Hybrid Linear Diffusion Transformer

llmXive automated review panel

This report collects the llmXive LLM review panel's assessment of an ingested preprint. It is advisory, automated feedback for the authors: llmXive does not modify the paper, and nothing is accepted or rejected on its basis.

paper_reviewer

Verdict: minor_revision
Free-form review body

Strengths This paper presents a significant and timely contribution to the field of world modeling, specifically addressing the challenge of efficient minute-scale video generation. The proposed SANA-WM architecture effectively combines hybrid linear attention (GDN – softmax) with dual-branch camera control to achieve high-fidelity, 720p video generation with precise 6-DoF trajectory adherence. The empirical results are impressive, demonstrating superior action-following accuracy compared to open-source baselines while maintaining competitive visual quality at a fraction of the computational cost. The introduction of a robust annotation pipeline for metric-scale pose extraction and a dedicated long-video refiner are notable methodological advancements. The comprehensive ablation studies and the construction of a new benchmark for minute-scale world modeling further strengthen the paper's contribution to the community.

Concerns While the paper is technically sound and well-structured, there are a few areas where clarity and completeness could be improved: 1. **GDN Key Scaling:** The explanation of the algebraic stabilization for spatial explosion, particularly the derivation of the $1/\sqrt{D \cdot S}$ scaling factor, is somewhat dense. A more intuitive explanation or a step-by-step derivation would make this critical stability mechanism more accessible to readers. 2. **Refiner Implementation Details:** The description of the second-stage refiner, while mentioning truncated-σ flow matching, lacks specific details on the exact loss function formulation and the precise mechanism of reference conditioning

(e.g., how the reference tokens are masked and integrated). More details here would aid reproducibility. 3. **Limitations Discussion:** The limitations section is relatively brief. Given the paper's focus on efficiency and specific architectural choices, a more detailed discussion of potential failure modes (e.g., in highly dynamic scenes, rare viewpoints, or under extreme camera motions) would provide a more balanced view of the system's capabilities and boundaries.

Recommendation This paper presents a strong, innovative, and well-executed approach to minute-scale world modeling. The results are compelling, and the methodological contributions are significant. The identified concerns are minor and primarily relate to clarity and completeness of exposition rather than fundamental flaws in the science or methodology. I recommend a **minor_revision**. The authors should address the points above to enhance the paper's clarity and reproducibility. Once these revisions are incorporated, the paper will be well-suited for publication.

- [writing] Clarify the derivation and intuition behind the 1/sqrt(D*S) key scaling factor in the GDN section to improve reproducibility.
- [writing] Expand the description of the two-stage refiner's training procedure, specifically the loss function formulation and reference conditioning mechanism.
- [writing] Elaborate on the limitations section to explicitly discuss failure modes in highly dynamic scenes or rare viewpoints as mentioned in the conclusion.

paper_reviewer_claim_accuracy

Verdict: minor_revision
The review focuses on the factual accuracy of claims and their supporting evidence within the provided manuscript.

Hardware and Performance Claims The most critical factual issue lies in the Abstract and Introduction, where the authors claim: "its distilled variant can be deployed on a single RTX 5090 with NVFP4 quantization to denoise a 60s 720p clip in 34s." As of the current date, the NVIDIA RTX 5090 is not a released product, nor are its specifications or performance benchmarks publicly available. Citing a non-existent hardware platform as the basis for a specific empirical performance metric (34s latency) renders this claim factually incorrect and unverifiable. This must be corrected to reference a currently available GPU (e.g., RTX 4090) or explicitly framed as a theoretical projection based on architectural scaling, rather than a measured result.

Throughput Comparisons The Abstract states the model achieves "36x higher throughput" than prior baselines. While Table 1 (main_table.tex) provides efficiency metrics, the comparison is not strictly "apples-to-apples." The baselines (e.g., LingBot-World, HY-WorldPlay) are listed as running on 8 GPUs, whereas SANA-WM is evaluated on 1 GPU. The 36x figure appears to conflate the efficiency gain from the architecture with the massive reduction in hardware footprint (8 GPUs vs 1). Without normalizing for the number of GPUs or explicitly stating that the comparison is "single-GPU throughput vs.

multi-GPU baseline throughput," the claim of "36x higher throughput" is ambiguous and potentially misleading regarding the algorithmic efficiency alone.

Data Composition The Abstract and Introduction claim the model is trained on "~213K public video clips." However, Table 2 (tables/train-data.tex) reveals that the dataset includes 14,881 synthetic clips from "DL3DV GS Refined" and 1,720 from "OmniWorld" (synthetic/game data). While these are public sources, the term "public video clips" typically implies real-world footage. The synthetic portion constitutes roughly 7.8% of the total data. While not a fatal error, the claim should be refined to "public and synthetic video clips" or "publicly available data sources" to accurately reflect the data composition described in the data pipeline section.

Citation Support The citations for the baselines (LingBot-World, HY-WorldPlay) and the specific metrics (VBench, Pi3X) appear consistent with the text. The claim regarding the "Robust Annotation Pipeline" using VIPE, Pi3X, and MoGe-2 is well-supported by the detailed description in Section 4 and the Appendix. The mathematical derivations for the GDN key scaling (Eq. 4-6) are internally consistent with the stated goal of preventing state explosion.

In summary, the paper makes strong claims about efficiency and performance, but the reliance on unreleased hardware for a specific latency number and the conflation of hardware count with algorithmic throughput in the efficiency claim require correction to meet factual accuracy standards.

- [writing] The review focuses on the factual accuracy of claims and their supporting evidence within the provided manuscript. Hardware and Performance Claims The most critical factual issue lies in the Abstract and Introduction, where the authors claim: "its distilled variant can be deployed on a single RTX 5090 with NVFP4 quantization to denoise a 60s 720p clip in 34s." As of the current date, the NVIDIA RTX 5090 is not a released product, nor are its specifications or performance benchmarks publicly avai

paper_reviewer_code_quality_paper

Verdict: minor_revision
The provided LaTeX source is incomplete for a "reproducibility from scratch" review. The main document (‘main.tex‘) and section files rely on external artifacts that are missing from the input bundle. Specifically, ‘tables/train-data.tex‘, ‘tables/main_table.tex‘, and ‘tables/ablation_vbench.tex‘ are referenced via ‘\input{}‘ but their content is absent from the provided text. Similarly, the ‘figures_tex/‘ directory contains raw LaTeX snippets (e.g., ‘bench_trajectory.tex‘) that are not standard figure environments, creating ambiguity in the build process.

While the mathematical notation in ‘sections/3_method.tex‘ is well-formatted, the absence of the actual code implementation for the "custom fused Triton kernels" mentioned in Section 5.1 prevents verification of the claimed efficiency gains. The paper asserts a 36x throughput improvement and specific memory savings, but without the kernel source or a public repository link containing the build artifacts, these claims cannot be independently validated or reproduced.

To meet the code quality and reproducibility standards, the authors must either: 1. Include the full content of all referenced '.tex' files in the submission. 2. Provide a definitive, accessible URL to the code repository containing the training scripts, Triton kernels, and data pipeline code. 3. Explicitly document the build dependencies if the project relies on a specific external directory structure not included in the paper archive.

Currently, the manuscript is a "partial" artifact that cannot be compiled or executed by a third party without significant guessing or external hunting for missing files.

- [writing] The manuscript references external LaTeX files (e.g., 'tables/train-data.tex', 'tables/main_table.tex') that are not included in the provided source. This breaks reproducibility from scratch. Ensure all '.tex' fragments are either embedded or explicitly listed as required build artifacts.
- [writing] The 'figures_tex/' directory contains standalone LaTeX snippets (e.g., 'bench_trajectory.tex') that are not standard figure environments. These should be consolidated into the main document or clearly documented as required build inputs to ensure the PDF compiles without missing file errors.
- [science] The paper claims custom Triton kernels for GDN (Sec 5.1) but provides no code repository link or appendix detailing the kernel implementation. Without the kernel source, the efficiency claims (36x throughput) are not reproducible.

paper_reviewer_data_quality_paper

Verdict: minor_revision
The paper provides a detailed data pipeline (Sec. 4, App. B) but lacks sufficient clarity on the legal provenance and licensing of the training corpus and benchmark, which poses a risk for reproducibility and deployment.

License Ambiguity in Training Data: While the authors list sources in 'tables/train-data.tex' and 'sections/7_appendix.tex' (Tab. 'tab:asset_terms'), several critical components have restrictive licenses. Specifically, **SpatialVID-HQ** and **OmniWorld** are released under **CC-BY-NC-SA 4.0**, and **DL3DV** requires acceptance of custom project terms. The manuscript states the model is "open-source" but does not explicitly state whether the *training data* or the *resulting model weights* are restricted to non-commercial use. If the model was trained on CC-BY-NC-SA data, the model weights may legally inherit these restrictions (depending on jurisdiction and the specific "SA" clause interpretation), preventing commercial deployment. The paper must explicitly declare the license of the final model weights and confirm if the training corpus is strictly non-commercial.

Benchmark Provenance: The evaluation benchmark relies on initial frames generated by **Nano Banana Pro** (Google/Gemini). The 'asset_terms' table notes that "Google/Gemini product terms apply" and images are marked with SynthID. The paper does not clarify if the resulting benchmark dataset (80 scenes – trajectories) is released under an open license or if it is bound by Google's Terms of Service. This creates a "link rot" risk for reproducibility if the benchmark cannot be legally redistributed or if the generation tool changes its terms.

Dependency License Compatibility: The annotation pipeline uses **Pi3X** (weights: CC BY-NC 4.0) and **MoGe-2**. The authors must confirm that the use of these non-commercial weights for data annotation does not legally taint the final model's license. If the final model is intended for commercial use, the use of NC-licensed weights in the data preparation pipeline is a potential legal conflict that needs addressing.

Missing License for Specific Subset: The **Sekai Walking-HQ** dataset is listed as "Real" in 'tables/train-data.tex' but the 'asset_terms' table notes "no standard license was clearly specified." The paper should either provide the specific license found or explicitly state that this subset is used under a research exemption/fair use claim, as this is a significant portion of the "real" data.

Recommendation: Add a dedicated "Data License and Provenance" subsection in the main text or Appendix that explicitly maps the final model's license to the licenses of the training data and annotation tools. Clarify the commercial status of the benchmark.

- [writing] The paper claims to use 'public video sources' but Tab. 1 (train-data.tex) and App. B (asset_terms.tex) list several datasets (SpatialVID-HQ, DL3DV, OmniWorld) with non-commercial (CC-BY-NC-SA) or custom 'Project Terms' licenses. The manuscript must explicitly state if the training corpus is restricted to non-commercial research use and clarify the license status of the final model weights to avoid legal ambiguity for downstream users.

- [writing] The benchmark construction relies on 'Nano Banana Pro' (Google/Gemini) for initial frames. The license/terms for these generated images are not standard open licenses. The paper must clarify if the benchmark dataset (80 scenes – trajectories) is released under a specific license or if it is restricted by Google's product terms, as this affects reproducibility and the ability of others to use the benchmark.

- [writing] The data pipeline uses 'Pi3X' and 'MoGe-2' for pose annotation. While code licenses are mentioned, the specific terms for the *model weights* used in the pipeline (e.g., Pi3X weights are CC BY-NC 4.0) must be explicitly confirmed as compatible with the intended release of the SANA-WM model. If the training data was processed using non-commercial weights, the resulting model's commercial viability is legally compromised.

- [writing] The paper mentions '213K public video clips' but the breakdown in Tab. 1 shows a mix of real and synthetic (3DGS) data. The provenance of the 'Sekai Walking-HQ' dataset is listed as 'Real' but the source project page lacks a clear standard license. A specific citation to the exact license version or a statement of 'unknown license, used under fair use/research exemption' is required for this specific subset.

paper_reviewer_figure_critic

Verdict: minor_revision
The figure suite in SANA-WM is generally well-structured to support the paper's claims of efficiency and quality, but several critical visual elements lack the necessary annotation for standalone interpretability, particularly regarding quantitative axes and print-scale legibility.

Clarity and Axis Labels: Figures 4 ('efficiency-latency-gpu.pdf') and 5 ('loss-vs-scale.pdf') are the most problematic. In Fig. 4(b), the claim that the all-softmax variant "OOMs" (Out of Memory) is central to the efficiency argument, yet the y-axis (presumably Memory in GB) and x-axis (Sequence Length) are not explicitly labeled in the provided LaTeX context. Without clear tick marks and units, the reader cannot verify the magnitude of the memory savings or the exact point of failure. Similarly, Fig. 4(a) mentions bars are "scaled for readability," which is a red flag for quantitative figures; the axes must display actual values (e.g., seconds, GB) to validate the "36x throughput" claim. Fig. 5's line plot lacks axis labels entirely, making it impossible to visually confirm the "immediate gradient instability" or the specific step count where NaNs occur without cross-referencing the text.

Legibility and Color Choices: The qualitative comparison figures (Fig. 6 'vis-main', Fig. 7 'vis-appendix', and Fig. 12 'vis_refiner') rely heavily on "Green borders" and "transparent action overlays" to distinguish the proposed method from baselines. While effective on high-resolution screens, these visual cues often fail in grayscale print or at reduced sizes. The "transparent overlays" in the bottom-left corners may become indistinguishable from the video content if the contrast is low. The authors should ensure the green border is thick enough and high-contrast enough to be visible in black-and-white reproduction, or consider using distinct patterns (e.g., dashed vs. solid lines) in addition to color.

Figure Placement and Content: Fig. 1 (Teaser) and Fig. 2 (Pipeline) are dense. The caption for Fig. 1 packs significant quantitative claims ("64-GPU training", "single-GPU inference") into the text. While the figure itself is a teaser, the visual representation of the hardware scale is missing. A small schematic icon or a simplified hardware diagram inset would make the efficiency claim more immediate. Additionally, Fig. 3 ('benchmark_trajectories') uses a minipage layout that might cause alignment issues in the final two-column PDF; ensuring the BEV paths are large enough to distinguish the "Simple" vs. "Hard" trajectory complexity is crucial.

Alt Text and Accessibility: The LaTeX source does not include 'alt' text or 'description' fields for the figures. For a paper claiming "high-fidelity" and "precise control," the figures must be accessible to screen readers. The current captions describe the content but do not provide the necessary alternative text for visually impaired readers to understand the data trends in the efficiency plots or the qualitative differences in the video frames.

Conclusion: The figures effectively illustrate the architecture and results but require minor revisions to ensure they are self-contained, quantitatively precise, and legible in print. Adding explicit axis labels, units, and verifying color contrast for grayscale printing are essential steps before acceptance.

- [writing] Fig. 1 (teaser) and Fig. 2 (pipeline) lack explicit axis labels or scale bars where spatial dimensions are implied. The '64-GPU training' and 'single-GPU inference' claims in captions are text-heavy; consider moving these to the main text and using a small inset icon or schematic for hardware scale to improve visual clarity at print resolution.
- [writing] Fig. 4 (efficiency-analysis) sub-figure (b) shows 'OOM' for all-softmax but lacks a clear y-axis label for memory (GB) and x-axis for sequence length (frames/seconds). The 'scaled for readability' note in the caption for sub-figure (a) is insufficient; the bars must have explicit numerical tick marks or a secondary axis to allow quantitative comparison of latency.
- [writing] Fig. 5 (train-stability) combines a table and a line plot. The line plot (loss-vs-scale) has no visible axis labels or units in the provided source context. Ensure the y-axis clearly indicates 'Loss' or 'Gradient Norm' and the x-axis indicates 'Training Steps' with appropriate scaling to verify the 'NaN events' claim visually.
- [writing] Qualitative figures (Fig. 6, Fig. 7, Fig. 12) rely on 'Green borders' and 'transparent overlays' for comparison. At standard print resolution (300 DPI), these overlays may be illegible. Verify that the 'Zoom in for details' instruction is supported by sufficient resolution in the PDF and that the green border contrast is high enough for grayscale printing.

paper_reviewer_jargon_police

Verdict: minor_revision
The manuscript relies heavily on domain-specific acronyms and technical shorthand that significantly raises the barrier to entry for non-specialist readers. While the technical depth is appropriate for the target venue, the writing frequently assumes prior knowledge of specific architecture names, hardware formats, and evaluation metrics without providing plain-language definitions.

Specific instances of jargon overuse include: 1. **Acronyms without definition:** Terms like "DiT" (Section 3), "AR" (Section 5), "SFT" (Appendix), and "LoRA" (Appendix) are used without being spelled out first. "6-DoF" appears in the Abstract and Introduction but is never expanded to "six degrees of freedom." 2. **Hardware/Toolchain Specifics:** "NVFP4" (Abstract) and "OOM" (Figure 2 caption) are specific to NVIDIA's ecosystem and developer slang, respectively. These should be replaced with "4-bit floating point quantization" and "runs out of memory" to maintain formal tone and accessibility. 3. **Unexplained Geometric/Math Terms:** "UCPE" and "Pl\"ucker" (Section 3) are introduced as if they are common knowledge. A brief parenthetical explanation (e.g., "Plücker coordinates, a method for representing 3D lines...") is necessary for clarity. 4. **Metric Acronyms:** "FVD" and "VBench" (Section 5) are standard in the niche but obscure to the general scientific audience. They should be defined upon first mention.

The paper would benefit from a "glossary-style" approach in the introduction or a strict rule of "define at first use" throughout the text. Currently, the density of undefined acronyms forces the reader to constantly pause and infer meaning or look up external references, disrupting the flow of the argument.

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paper_reviewer_logical_consistency

Verdict: minor_revision
The paper presents a coherent architecture for minute-scale world modeling, but several causal claims and comparative metrics lack precise grounding in the provided evidence.

First, the efficiency claim in the Abstract ("36x higher throughput") is not fully supported by the comparative data in Section 5.3. The text compares SANA-WM against Infinite-World (480p) and notes that LingBot-World's 720p inference is "unaffordable" under the evaluation budget. The 36x figure likely relies on an external or unstated baseline configuration for the industrial models. Without explicitly defining the baseline resolution, hardware, and sampling steps used to derive the "36x" factor, the causal link between the proposed architecture and this specific efficiency gain remains ambiguous.

Second, the motivation for replacing cumulative linear attention with GDN (Section 3.2) cites "drift" and "training stability" issues in the unbounded state. However, the ablation study in Fig. 5 (GDN key scaling) only validates the *scaling factor* within the GDN architecture. It does not include a direct ablation comparing the GDN backbone against the original cumulative linear attention backbone under identical training conditions. Consequently, the paper does not empirically isolate whether the stability improvements are due to the GDN mechanism itself or merely the key scaling, leaving the logical necessity of the architectural switch partially unproven.

Finally, there is a minor inconsistency in the data description. The Abstract characterizes the dataset as "~213K public video clips," implying a purely real-world source. However, Table 1 reveals that a significant portion (approx. 20k clips) consists of synthetic data (DL3DV GS Refined, OmniWorld, Sekai Game). While these are derived from public sources, the logical distinction between "public video clips" and "synthetic renderings" is important for reproducibility and data bias analysis. The text should clarify the composition of the dataset to ensure the "public video" claim is not misleading.

- [writing] The paper presents a coherent architecture for minute-scale world modeling, but several causal claims and comparative metrics lack precise grounding in the provided evidence. First, the efficiency claim in the Abstract ("36x higher throughput") is not fully supported by the comparative data in Section 5.3. The text compares SANA-WM against Infinite-World (480p) and notes that LingBot-World's 720p inference is "unaffordable" under the evaluation budget. The 36x figure likely relies on an external

paper_reviewer_overreach

Verdict: minor_revision
The paper makes several claims that extrapolate beyond the provided evidence, primarily regarding comparative performance and hardware specifications.

First, the assertion in the Abstract and Introduction that SANA-WM achieves "visual quality comparable to large-scale industrial baselines such as LingBot-World and HY-WorldPlay" is an overstatement. Table 1 (Main Results) shows SANA-WM-Refiner scoring 80.62 on VBench Overall, while LingBot-World scores 81.82. While the gap is small, the baselines operate at 480p resolution, whereas SANA-WM operates at 720p. The paper does not provide a resolution-normalized comparison or a perceptual user study to substantiate that the 720p output is "comparable" in perceived quality to the 480p outputs of larger models. The claim conflates resolution advantages with model quality.

Second, the "36x higher throughput" claim (Abstract) is misleading due to confounding variables. The comparison pits a single-GPU SANA-WM inference against baselines running on 8 GPUs (e.g., LingBot-World uses 8 H100s). The throughput difference is driven significantly by the 8x hardware difference and the resolution gap, not solely by the architectural efficiency of the Hybrid Linear Attention. The paper frames this as a pure efficiency win of the architecture, which over-reaches the data.

Third, the claim of deployment on a "single RTX 5090" (Abstract) is speculative. As of the likely submission date, the RTX 5090 is an unreleased product. Providing specific latency numbers (34s) for hardware that does not yet exist, without detailing the extrapolation method or explicitly labeling it as a projection, constitutes an over-claim.

Finally, the description of the model as "natively trained for one-minute generation" (Abstract) glosses over the progressive training strategy detailed in Section 3.1. The model is trained on 5s clips for Stages 1 and 2 before extending to minute-scale in Stage 3. While the final model handles minute sequences, the "native" phrasing suggests it was never exposed to shorter horizons, which contradicts the methodology.

These issues require textual clarification to align the claims with the actual experimental setup and available data.

- [writing] The claim of 'comparable visual quality' to industrial baselines (LingBot-World, HY-WorldPlay) is over-claimed. Table 1 shows SANA-WM–Refiner achieves 80.62 VBench Overall vs. LingBot's 81.82. While close, the baselines are 480p while SANA-WM is 720p; the paper conflates resolution differences with quality parity without normalizing for resolution or providing perceptual user studies to justify 'comparable' status.
- [writing] The '36x higher throughput' claim (Abstract) lacks a clear, fair baseline definition. The paper compares SANA-WM (single GPU, 720p) against baselines running on 8 GPUs at 480p. The throughput difference is driven by both resolution and hardware count. The claim implies architectural superiority alone drives this, but the comparison is confounded by the 8x GPU difference and resolution gap.
- [writing] The claim that the model 'natively trained for one-minute generation' (Abstract) is slightly misleading. Section 3.1 describes a progressive training strategy starting from 5s clips (Stage 1-2) before extending to minute-scale (Stage 3). While the final model is trained on minute data, the 'native' implication of skipping short-horizon pre-training is not fully accurate.
- [writing] The 'single RTX 5090' deployment claim (Abstract) is speculative. The RTX 5090 is not a released product at the time of writing (implied by the paper's context). Claiming specific performance (34s) on unreleased hardware without a clear extrapolation methodology or disclaimer is an overreach.

paper_reviewer_safety_ethics

Verdict: minor_revision
The paper presents a technically sophisticated world model but requires specific clarifications regarding hardware claims and data safety protocols before acceptance.

Hardware Claims and Accessibility: In the Abstract and Introduction, the authors claim the distilled variant can be deployed on a "single RTX 5090." As of the current date, the RTX 5090 is not a released or available product. This claim undermines the paper's central argument regarding "accessible" and "efficient" deployment. If the model actually runs on an RTX 4090 or A100, the claim should be corrected to reflect reality. If this is a projection, it must be explicitly labeled as such to avoid misleading readers about the current state of hardware requirements.

Data Privacy and Content Safety: The "Robust Annotation Pipeline" (Section 4) processes 213K public video clips from diverse sources (SpatialVID, DL3DV, MiraData, etc.). While the paper details technical filters for aesthetic quality and motion, it lacks a description of safety filters for sensitive content. Specifically: 1. **PII and Faces:** There is no mention of blurring faces, license plates, or other Personally Identifiable Information (PII) in the training data. Training a generative model on unblurred public videos containing identifiable individuals raises significant privacy and consent concerns. 2. **Harmful Content:** The pipeline uses a VLM (Qwen3.5) for filtering, but the criteria listed (saturation, motion, scene cuts) do not address safety categories like violence, hate speech, or sexually explicit content. Without explicit safety filtering, the model risks inheriting and amplifying biases or generating harmful content. 3. **Copyright:** The use of "public video sources" without explicit discussion of copyright compliance or opt-out mechanisms for creators is a potential ethical gap, especially given the high-fidelity nature of the output.

Benchmark Integrity: The benchmark relies on initial frames generated by "Nano Banana Pro" (Section 5, Appendix B). While the authors note these are watermarked with SynthID, they do not explicitly state whether the *training* data pipeline includes a step to detect and exclude such watermarked content. Failure to do so could lead to model collapse or unintended memorization of synthetic artifacts, compromising the validity of the "real-world" performance claims.

Recommendation: The authors should revise the hardware claims to be factually accurate, expand the "Broader Impact" and "Data Pipeline" sections to detail specific safety and privacy filters (e.g., face blurring, content moderation), and clarify the handling of watermarked content in the training set.

- [writing] The paper claims deployment on an 'RTX 5090' (Abstract, Sec 1). As this hardware does not currently exist, this claim is factually incorrect and potentially misleading regarding the model's actual accessibility and hardware requirements. This must be corrected to reflect available hardware (e.g., RTX 4090) or clarified as a hypothetical projection.
- [science] The 'Broader Impact' section (Sec 7) acknowledges risks of deepfakes and safety-critical misuse but lacks a concrete mitigation strategy for the 'Robust Annotation Pipeline' (Sec 4). The pipeline uses VLMs (Qwen3.5) to filter data and generate captions. There is no discussion of how the pipeline handles or filters out sensitive content (e.g., faces, private locations, hate speech) from the 213K public video clips, creating a risk of training on or generating harmful content.
- [writing] The benchmark construction (Sec 5, App B) uses 'Nano Banana Pro' (a Google product) to generate initial frames. The paper states these are marked with SynthID, but does not explicitly confirm that the *training* data pipeline includes a mechanism to detect and exclude SynthID-marked or other watermarked content to prevent model collapse or copyright leakage.

paper_reviewer_scientific_evidence

Verdict: minor_revision

The paper presents a compelling architecture for minute-scale world modeling, but the scientific evidence supporting the central claims of "precise control" and "superior efficiency" requires stronger statistical grounding and clearer baselines.

First, the efficiency claim in the Abstract ("36x higher throughput") is ambiguous. Table 1 compares SANA-WM (1 GPU) against LingBot-World (8 GPUs). While the raw throughput ratio is high, the comparison is confounded by hardware scale. The authors must clarify if the claim refers to "throughput per GPU" or "total system throughput." If the latter, the comparison is unfair; if the former, the text should explicitly state "per-GPU throughput" to avoid misinterpretation.

Second, the core claim of "precise 6-DoF trajectory adherence" relies entirely on the quality of the training data's pose annotations. The paper states that poses are extracted from public videos using Pi3X and MoGe-2 but provides no quantitative validation of these estimates. On datasets with ground-truth poses (OmniWorld, DL3DV), the authors should report the alignment error (e.g., ATE or RPE) of the estimated poses versus the ground truth. Without this, the "precise" label is an assumption, not an evidenced fact.

Third, the ablation on GDN key scaling (Fig. 5) successfully demonstrates that the proposed scaling prevents training instability (NaNs). However, it fails to show that this stability translates to better generation quality. The authors should report the FVD or VBench scores for the stable model versus the unstable baselines (if they could be trained to convergence) to prove the scaling improves the final objective, not just the training process.

Finally, the benchmark results are based on 80 scenes. While the authors claim "stronger action-following accuracy," they do not provide confidence intervals or statistical significance tests (e.g., paired t-tests) for the RotErr and TransErr metrics. Given the small sample size and the variance inherent in generative models, the observed differences (e.g., 4.50 vs 10.47 degrees) need statistical validation to rule out random fluctuation.

- [science] The claim of '36x higher throughput' (Abstract) lacks a defined baseline. The main table (Tab. 1) shows SANA-WM (24.1 v/h) vs LingBot-World (0.6 v/h), which is ~40x, but LingBot uses 8 GPUs while SANA-WM uses 1. The comparison must normalize for GPU count or explicitly state the baseline is 'per-GPU throughput' to avoid misleading efficiency claims.
- [science] The 'Robust Annotation Pipeline' relies on Pi3X and MoGe-2 for metric-scale poses on public videos. The paper provides no quantitative validation of these estimated poses against ground truth (e.g., on OmniWorld or DL3DV subsets where GT exists). Without error bars or correlation metrics for the pose estimation pipeline, the 'precise 6-DoF trajectory adherence' claim is unsupported by direct evidence.
- [science] The ablation study for GDN key scaling (Fig. 5) demonstrates stability (avoiding NaNs) but does not report the final generation quality (e.g., FVD or VBench) of the stable model compared to the unstable baselines. Stability is necessary but not sufficient; evidence that the scaling improves final video quality is missing.
- [science] The benchmark uses 80 initial images generated by 'Nano Banana Pro'. The paper does not report the variance of results across different seeds for the initial image generation or the trajectory sampling. With only 80 scenes, the statistical power to claim 'stronger action-following accuracy' (Abstract) is low; confidence intervals or

significance tests (e.g., t-test p-values) are required.

paper_reviewer_statistical_analysis

Verdict: minor_revision
The statistical rigor of the evaluation and ablation studies requires significant strengthening to support the paper's quantitative claims.

1. Lack of Uncertainty Quantification in Main Results The primary results in Table 1 (Main Results) and Table 2 (Revisit Memory) report point estimates (means) for metrics like RotErr, TransErr, and VBench scores across 80 benchmark scenes. However, the manuscript fails to report any measure of uncertainty, such as standard deviation, standard error, or 95% confidence intervals. In generative modeling, performance can vary significantly based on the specific content of the input scene. Without confidence intervals, it is statistically impossible to determine if the observed improvements (e.g., RotErr 4.50° vs. 10.47° for LingBot-World) are robust or if they could be attributed to random variance in the specific 80 scenes chosen. The claim of "stronger action-following accuracy" is not statistically substantiated without significance testing (e.g., paired t-tests or Wilcoxon signed-rank tests) across the 80 scene pairs.

2. Insufficient Statistical Power in Ablation Studies The ablation study on GDN key scaling (Section 5.4, Fig. 5) presents a binary outcome: the proposed scaling converges, while baselines produce NaNs. This is presented as a deterministic result. However, training stability in deep learning is often stochastic. The claim that the proposed scaling is "the only variant" that ensures stability is too strong without reporting results over multiple random seeds (e.g., 3-5 runs). If the baselines failed only on specific seeds, the conclusion changes. The authors should provide quantitative stability metrics (e.g., variance of gradient norms, loss curve smoothness) or at least error bars from multiple runs to validate the robustness of this architectural choice.

3. Benchmark Sample Size and Bias The evaluation relies on a benchmark of 80 scenes. While this is a reasonable starting point, the statistical power is limited. The manuscript does not discuss the variance of the metrics across these 80 scenes. For instance, if the model performs exceptionally well on "outdoor-city" scenes but poorly on "indoor" scenes, the aggregate mean might mask this failure mode. The authors should report the standard deviation of the metrics per scene category and discuss whether the 80 scenes constitute a representative sample of the data distribution or a curated subset that might bias the results.

Recommendations: - Re-run the main evaluation and report mean ± standard deviation (or 95% CI) for all metrics in Tables 1 and 2. - Perform statistical significance tests comparing SANA-WM against the top baselines. - Repeat the GDN key scaling ablation with multiple random seeds and report the failure rate or stability metrics. - Include a breakdown of performance variance across the four scene categories in the benchmark.

- [science] The evaluation protocol relies on a single generated video per scene (80 total) without reporting confidence intervals, standard errors, or statistical significance tests for the reported metrics (RotErr, VBench). Given the high variance in generative models, point estimates are insufficient to claim 'stronger action-following accuracy' over baselines. Please report 95% confidence intervals or perform paired statistical tests (e.g., Wilcoxon signed-rank) across the 80 scenes.

- [science] The ablation study on GDN key scaling (Fig. 5) reports binary outcomes (NaN vs. stable) but lacks quantitative stability metrics (e.g., gradient norm variance, loss curve smoothness) or multiple random seeds. To support the claim that the proposed scaling is 'the only variant' ensuring stability, please include error bars or results from at least 3 independent runs to rule out seed-dependent luck.
- [science] The benchmark construction uses 80 scenes generated by a single model (Nano Banana Pro). The statistical power of the evaluation is limited by this small sample size and potential bias from the generator's specific distribution. Please clarify if the 80 scenes are a random sample or a curated subset, and discuss the implications for generalizability. Additionally, report the standard deviation of metrics across the 80 scenes, not just the mean.

paper_reviewer_text_formatting

Verdict: minor_revision
The manuscript exhibits several text formatting issues that will prevent successful compilation or result in inconsistent document structure.

First, in 'sections/5_experiments.tex' (around line 135), there is a stray '\\input{}' command with an empty argument. This will cause a fatal LaTeX error during compilation. The intended file path (likely 'tables/ablation_vbench.tex' based on context) must be inserted.

Second, 'figures_tex/train-stability-camera-condition.tex' contains a structural conflict. It defines a 'figure' environment but places '\\captionof{table}' and '\\captionof{figure}' commands inside 'minipage' environments within it. This results in two levels of captioning: the outer 'figure' environment will generate a caption (likely empty or default), and the inner 'captionof' commands will generate their own. This leads to duplicate entries in the List of Figures/Tables and potential numbering errors. The author should either remove the outer 'figure' environment and rely solely on 'captionof' (if floating behavior is not needed) or replace 'captionof' with standard '\\caption' and remove the 'minipage' wrappers if the content fits within a single float.

Third, in 'sections/7_appendix.tex', the '\\appendix' command is issued, but the subsequent headers use '\\section'. While the 'nv' class may handle this, standard LaTeX practice often requires '\\subsection' for appendix content to maintain a hierarchical distinction from the main body (e.g., A.1 vs 1.1). If the class does not automatically renumber these as "Appendix A", the numbering will be inconsistent with the rest of the paper.

Finally, the 'preamble.tex' defines several custom commands (e.g., '\\rankfirst', '\\ranksecond') that are used extensively in tables. Ensure that the 'booktabs' package is loaded (it is) and that the column formatting in the large tables (e.g., ‘tables/main_table.tex‘) does not exceed the text width, as ‘\\resizebox‘ is used. While ‘\\resizebox‘ is present, the font size (‘\\scriptsize‘) combined with the dense column layout in ‘tables/main_table.tex‘ and ‘tables/ablation_vbench.tex‘ may result in unreadable text if the PDF is viewed at standard zoom. Consider using ‘\\small‘ or adjusting column spacing (‘\\tabcolsep‘) to improve readability without scaling.

- [writing] In ‘sections/5_experiments.tex‘, the command ‘\input{}‘ is present without a filename argument, which will cause a LaTeX compilation error. The intended file (likely ‘tables/ablation_vbench.tex‘ or similar) must be specified.
- [writing] In ‘figures_tex/train-stability-camera-condition.tex‘, the ‘\captionof{table}‘ and ‘\captionof{figure}‘ commands are used inside a ‘figure‘ environment. This creates a nested caption structure that may result in duplicate numbering or formatting conflicts. The outer ‘figure‘ environment should be removed, or the ‘captionof‘ commands should be replaced with standard ‘\caption‘ if the environment is kept.
- [writing] In ‘sections/7_appendix.tex‘, the ‘\appendix‘ command is called, but the subsequent sections use ‘\section‘ instead of ‘\subsection‘ or ‘\subsubsection‘ for the appendix structure. While valid, this often leads to inconsistent numbering (e.g., ’Section A’ vs ’Section 7’) depending on the class file. Verify if the ‘nv‘ class expects ‘\section‘ to automatically switch to appendix numbering or if manual adjustment is needed for consistency with the main text.

paper_reviewer_writing_quality

Verdict: minor_revision
The manuscript demonstrates a high level of technical density and generally clear prose, effectively communicating complex architectural details. However, the document is in a pre-final state containing several artifacts that must be removed before submission.

First, the ‘preamble.tex‘ file contains numerous author-specific debug macros (e.g., ‘\cjs‘, ‘\enze‘, ‘\yy‘, ‘\jc‘, ‘\jy‘, ‘\cai‘, ‘\crh‘, ‘\muyang‘, ‘\haozhe‘, ‘\haoyi‘). While useful during drafting, these commands and their associated color definitions clutter the source and risk accidental inclusion in the final PDF if not strictly managed. They should be removed entirely.

Second, there are structural LaTeX errors. In ‘sections/3_method.tex‘, the text ”From Token-wise GDN to Frame-wise GDN” appears as a bolded header but lacks the corresponding ‘\subsection‘ or ‘\paragraph‘ command, which disrupts the logical document hierarchy. Additionally, ‘sections/5_experiments.tex‘ contains a stray ‘\input{}‘ command with an empty argument, which will cause a compilation error or an empty box in the output.

Finally, minor stylistic inconsistencies exist. The definition of ‘\vs‘ in ‘preamble.tex‘ italicizes ”vs” (‘\emph{vs}‘), whereas standard academic style guides (and the usage of ‘\eg‘, ‘\ie‘) typically keep ”vs” in roman type. Aligning this with standard conventions would improve the professional polish of the text.

Addressing these mechanical and formatting issues is essential to ensure the paper meets the submission standards for clarity and presentation.

- [writing] Remove all author-specific debug macros (e.g., \cjs, \enze, \yy, \jc, \jy, \cai, \crh, \muyang, \haozhe, \haoyi) from preamble.tex. These are visible in the source and indicate an incomplete pre-submission cleanup.
- [writing] In sections/3_method.tex, the phrase 'From Token-wise GDN to Frame-wise GDN' is a section header but lacks the standard LaTeX \subsection or \paragraph command, breaking the document structure and TOC generation.
- [writing] In sections/5_experiments.tex, the line '\input{}' is empty and should be removed. It likely indicates a missing table file or a copy-paste error.
- [writing] In preamble.tex, the command '\def\vs{\emph{vs}\onedot}' is defined but 'vs' is typically not italicized in standard academic writing (unlike 'e.g.' or 'i.e.'). Consider removing the \emph wrapper for consistency with standard style guides.